

Kalshi Tmax Market Strategy — Research Log

Oscar Ovanger

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1 Data and Market Structure

1.1 Overview

This project evaluates trading strategies for Kalshi same-day maximum temperature (Tmax) prediction markets across 12 US cities. Each trading day, Kalshi offers a set of mutually exclusive 2°F-wide temperature buckets covering the plausible Tmax range. One bucket resolves to \$1 (YES wins); all others resolve to \$0. Settlement uses the official NWS Daily Climate Report (CLI) for each city’s ASOS station. The dataset contains 5-minute market snapshots from March 30 to June 14, 2026, restricted to the post-10:00 AM local time trading window. Table 1a summarises the city-level inventory.

1.2 Evaluation Design

The dataset is split along both time and location axes (Figure 1). Nine train cities are used for model development; three holdout cities (Denver, Miami, Minneapolis) are reserved for generalisation testing and are never used for parameter tuning. The time axis assigns the most recent 15 calendar dates to an out-of-sample (OOS) partition (time_holdout), with earlier dates forming the in-sample partition (threshold_opt). A fresh validation window (June 3–14, 2026) extends beyond the original time_holdout partition to test deployment readiness on the most recent market data. A final true_holdout partition (holdout cities × OOS dates) is never inspected until final reporting.

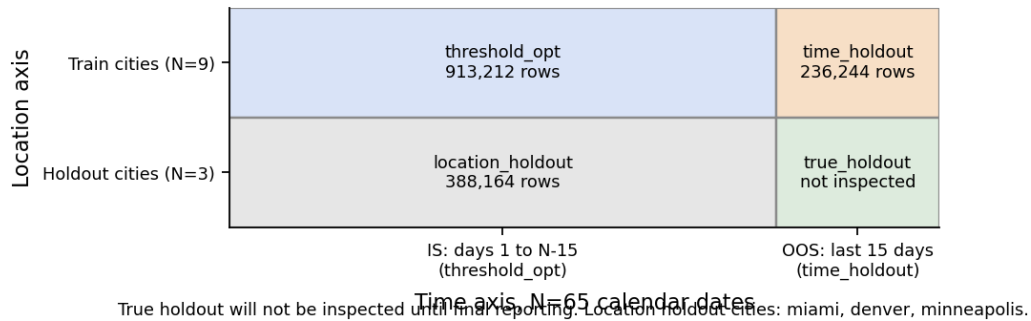


Figure 1

Table 1

(a) City inventory with N days, N snapshots, and date range.

City	N days	N snapshots	First date	Last date
austin	61	17607	2026-03-30	2026-06-20
chicago_midway	73	21727	2026-03-30	2026-06-20
denver†	61	17918	2026-03-30	2026-06-02
houston	77	22627	2026-03-30	2026-06-20
los_angeles	77	22598	2026-03-30	2026-06-20
miami†	55	16099	2026-03-30	2026-05-30
minneapolis†	64	18601	2026-03-30	2026-06-02
new_york_city	76	22370	2026-03-30	2026-06-20
oklahoma_city	78	22995	2026-03-30	2026-06-20
philadelphia	67	20003	2026-03-30	2026-06-20
phoenix	79	23316	2026-03-30	2026-06-20
san_francisco	78	22983	2026-03-30	2026-06-20

† Held-out city: not used for parameter tuning.

1.3 Fee Model

All profit and loss figures are reported net of Kalshi fees. The taker fee in cents is $\lceil 0.07 \cdot C \cdot P \cdot (1 - P) \rceil$ and the maker fee is $\lceil 0.0175 \cdot C \cdot P \cdot (1 - P) \rceil$, where C is the number of contracts and P is the YES mid price. The ceiling function creates a 1-cent minimum per trade; for positions under approximately 57 contracts, the fee is pinned at 1 cent regardless of size. Figure 2 shows the fee schedule for $C = 100$ contracts. A no-trade guardrail skips any bucket where the estimated edge ($p - c$) is less than twice the per-contract fee, and a price floor excludes buckets priced below \$0.15.

2 Market Baselines and Efficiency

2.1 Baseline Definitions

Seven market baselines test whether simple rules extract profit from the Tmax bucket structure. Entry timing uses a snapshot-stability rule: the first post-10:00 AM snapshot at which the modal bucket (highest YES mid price) has been unchanged for $k = 1$ consecutive 5-minute interval, which in practice fires immediately at market open. The implied-favorite baseline enters long on the modal bucket at this snapshot. The distribution-copy baseline allocates capital across all buckets proportional to fee-adjusted price, serving as a theoretical zero-edge benchmark. The sell-longshots baseline shorts any bucket crossing below \$0.10 after the stability snapshot, motivated by the documented favorite-longshot bias. The make-the-market baseline quotes one cent inside the modal bucket spread until a maker fill arrives. Three signal-threshold baselines enter when mode probability exceeds $t^* = 0.88$ (mode-prob), Shannon entropy falls below $H^* = 0.45$ (entropy), or mode probability momentum exceeds $d^* = 0.02$ over $w^* = 16$ snapshots (momentum). Threshold values were frozen on the in-sample partition.

Table 2

(a) IS and OOS statistics for all seven market baselines. IS uses threshold_opt (416 day-equiv); OOS uses time_holdout (108 day-equiv). No re-optimisation occurred between partitions.

Baseline	IS N	OOS N	IS SR	OOS SR	OOS CI	PSR(0)	Sortino	Max DD
implied favorite	412	107	-0.35	1.16	[-1.84, 4.16]	0.77	6.39	-3.79
distribution copy	412	107	-9.71	-6.05	[-9.15, -2.95]	0.00	-5.91	-4.58
sell longshots	388	97	-0.28	-0.60	[-3.60, 2.39]	0.34	-0.63	-3.54
make the market	411	107	-0.45	1.45	[-1.55, 4.45]	0.82	8.13	-3.79
mode prob threshold	170	50	-0.83	0.10	[-2.89, 3.09]	0.53	0.77	-1.40
entropy threshold	139	44	0.42	-0.49	[-3.48, 2.51]	0.37	-1.96	-1.81
momentum threshold	400	104	-0.52	1.36	[-1.63, 4.36]	0.81	6.98	-4.07

2.2 Intraday Market Dynamics

Figure 3 shows the average post-10 AM intraday mode probability across all train-city trading days, illustrating how the market concentrates toward a dominant bucket as the day progresses.

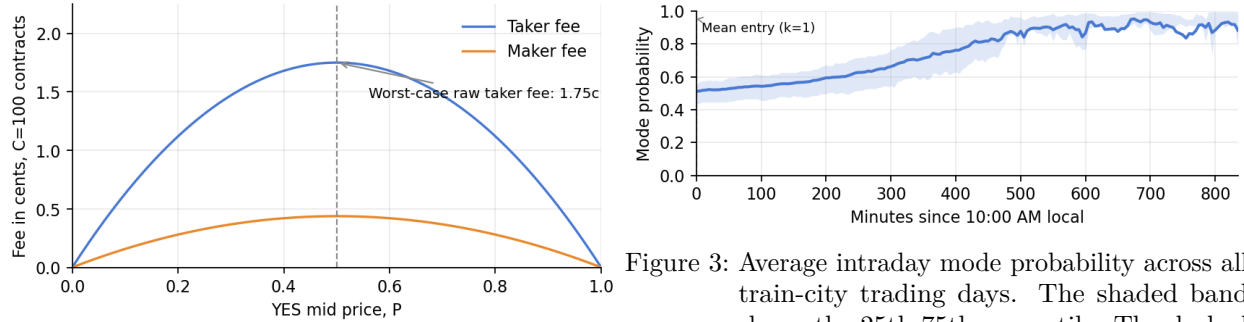


Figure 2: Kalshi fee schedule for $C=100$ contracts. Fees peak at $P=0.50$ and are substantially lower near the tails.

Figure 3: Average intraday mode probability across all train-city trading days. The shaded band shows the 25th-75th percentile. The dashed line marks the mean snapshot at which the stability rule fires.

2.3 Baseline Results

Table 2a reports IS and OOS statistics for all seven baselines. No baseline achieves a positive deflated Sharpe or meets the minimum track record length (MinTRL) required for statistical significance at the 5% level. The Spearman rank correlation between IS and OOS Sharpe is 0.036, indicating that IS rankings provide essentially no information about OOS performance. Three baselines show positive OOS point estimates (make-the-market 1.45, momentum 1.36, implied-favorite 1.16), but all 95% confidence intervals include zero. Furthermore, 138% of combined OOS PnL is concentrated in the top three trading days, suggesting these positive results are driven by a small number of favorable days rather than a persistent signal.

2.4 IS versus OOS Comparison

Figure 4 compares IS and OOS Sharpe across baselines; the pronounced rank inversion and wide OOS confidence intervals underscore the preliminary nature of any performance claim on 108 day-equivalents.

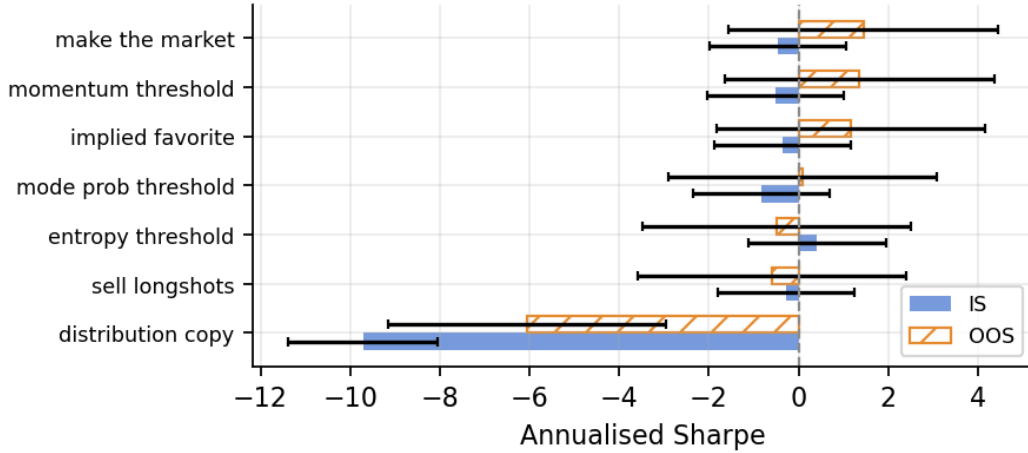


Figure 4

3 Forecast Models

3.1 Architecture

Track-B is a three-learner ensemble that predicts next-day maximum temperature in degrees Fahrenheit for each of the nine train cities. The ensemble combines three regressors chosen for complementary inductive biases. Ridge regression provides a regularised linear baseline, with the penalty parameter α selected from $\{0.01, 0.1, 1.0, 10.0, 100.0\}$ by minimising validation MAE. Huber regression ($\epsilon = 1.35$) fits a linear model robust to outliers, downweighting days where the residual exceeds 1.35 standard deviations; this guards against anomalous Tmax events (e.g. cold fronts, record heat) that would distort ordinary least-squares. LightGBM captures nonlinear feature interactions that the linear learners miss, trained with 500 estimators, maximum depth 4, learning rate 0.05, and early stopping on the validation set (patience 30 rounds). Both Ridge and Huber are fit on standardised features via scikit-learn pipelines.

The final prediction is the unweighted mean of the three base learners, rounded to the nearest integer degree Fahrenheit. Ensemble weights are not tuned: equal weighting avoids overfitting the combination to the validation set, which is a particular concern given that the validation window (2025) spans only one year. Models are trained on 2021–2024 data, validated on 2025, and tested on 2026 onward. The training and validation sets are strictly non-overlapping, and no test-period data is used at any stage of model fitting or selection.

The model ingests 25 candidate features across five groups: (1) nine ASOS morning observations (temperature, dewpoint, humidity, pressure, wind, cloud cover as of 10:00 AM), (2) nine calendar and temperature-lag features (day-of-year harmonics, lagged Tmax from 1 to 7 days, rolling means), (3) NWS MOS Tmax forecast issued the previous evening, (4) three GFS afternoon forecast fields (2-m temperature, dewpoint, cloud cover from the 00Z cycle), and (5) best-available NWP Tmax from ECMWF IFS (priority) or GFS seamless. Per-city feature selection on the training set drops columns with more than 20% missingness or absolute Pearson correlation below 0.05 with Tmax, retaining a mean of 22 features per city. A leakage audit confirmed that all GFS features use forecast hours ($fx > 0$) available before the 10:00 AM trading cutoff.

3.2 Bucket Probability Conversion

The integer-valued ensemble prediction $\hat{T}$ is converted to per-bucket probabilities via a Gaussian CDF assumption. For a RANGE bucket $[a, b]$: $P_k = \Phi((b - \hat{T})/\sigma) - \Phi((a - \hat{T})/\sigma)$. For LESS_THAN and

Table 3

(a) Track-B per-city training summary (train < 2025, val = 2025, test ≥ 2026).

City	N train	N val	N test	N feat	Val MAE	Test MAE	Test ±1°F	Sigma
austin	655	342	146	22	1.76	2.21	36.3%	2.12
chicago midway	714	361	154	23	1.69	1.87	53.2%	1.38
houston	669	338	153	22	1.30	1.42	64.7%	1.08
los angeles	699	360	151	23	1.26	1.54	54.3%	1.34
new york city	713	360	156	22	1.43	1.76	54.5%	1.34
oklahoma city	716	353	154	22	1.86	2.57	42.9%	1.77
philadelphia	714	359	149	23	1.67	2.33	30.9%	2.52
phoenix	653	344	138	20	1.40	1.29	67.4%	1.02
san francisco	709	359	146	22	1.93	2.00	40.4%	1.89

Table 4

(a) Track-B versus aligned Track-A/Track-J and raw NWS comparison on the 2026+ test window. Austin baseline is Track-J.

City	Track-A MAE	Track-B MAE	Improvement	Track-A ±1°F	Track-B ±1°F	Raw NWS MAE
austin	1.84	2.21	-0.37	46.4%	36.3%	6.27
chicago midway	2.85	1.87	0.98	30.7%	53.2%	8.36
houston	2.79	1.42	1.36	32.9%	64.7%	4.45
los angeles	1.80	1.54	0.25	54.4%	54.3%	3.28
new york city	3.65	1.76	1.89	25.3%	54.5%	6.59
oklahoma city	4.39	2.57	1.82	20.7%	42.9%	7.62
philadelphia	3.94	2.33	1.61	15.9%	30.9%	6.65
phoenix	1.82	1.29	0.54	55.0%	67.4%	3.47
san francisco	2.49	2.00	0.49	30.9%	40.4%	3.45

GREATER_THAN tail buckets, the corresponding one-sided CDF applies. The scale parameter σ is back-solved from the test-set $\pm 1^\circ\text{F}$ hit rate h using $h = 2\Phi(1/\sigma) - 1$. Cities with higher hit rates produce smaller σ and therefore more peaked (confident) bucket distributions. Table 3a reports the back-solved σ for each city; values range from 1.02°F (Phoenix, tightest) to 2.52°F (Philadelphia, widest), directly determining how aggressively the model concentrates probability on its predicted bucket and therefore how large the implied edge is when the model disagrees with the market.

3.3 Per-City Forecast Accuracy

Table 3a reports Track-B accuracy on the 2026 test window. Phoenix achieves the best performance (MAE 1.29°F , 67.4% $\pm 1^\circ\text{F}$ hit rate, $\sigma = 1.02^\circ\text{F}$), followed by Houston (1.42°F , 64.7% , $\sigma = 1.08^\circ\text{F}$). Five cities exceed 50% hit rate. Philadelphia is the weakest (30.9% , $\sigma = 2.52^\circ\text{F}$). Table 4a compares Track-B against aligned Track-A baselines (scored on the same 2026 dates) and raw NWS MOS forecasts. Track-B improves over Track-A for eight of nine cities, with the largest gains at New York City (1.89°F MAE reduction), Oklahoma City (1.82°F), and Philadelphia (1.61°F). Austin is the sole exception: the earlier Track-J model (which uses a richer 6-learner architecture with HRRR/RAP features) retains a 0.37°F advantage over the standardised Track-B pipeline.

3.4 Feature Importance

Figure 5 shows LightGBM split importances averaged across all nine cities. The best-available NWP forecast (nwp_tmax_best_f) and GFS afternoon temperature (gfs_t2m_afternoon) dominate, confirming that Track-B is primarily a bias-corrected NWP forecast. The 10:00 AM surface temperature (temp_10am) ranks

third, providing same-morning ground truth that the NWP models lack. No leakage features (resolved, settled, or bucket fields) appear in the retained sets. Figure 6 shows the per-city MAE comparison across Track-A/Track-J, Track-B, and raw NWS, with a 2.0°F reference line.

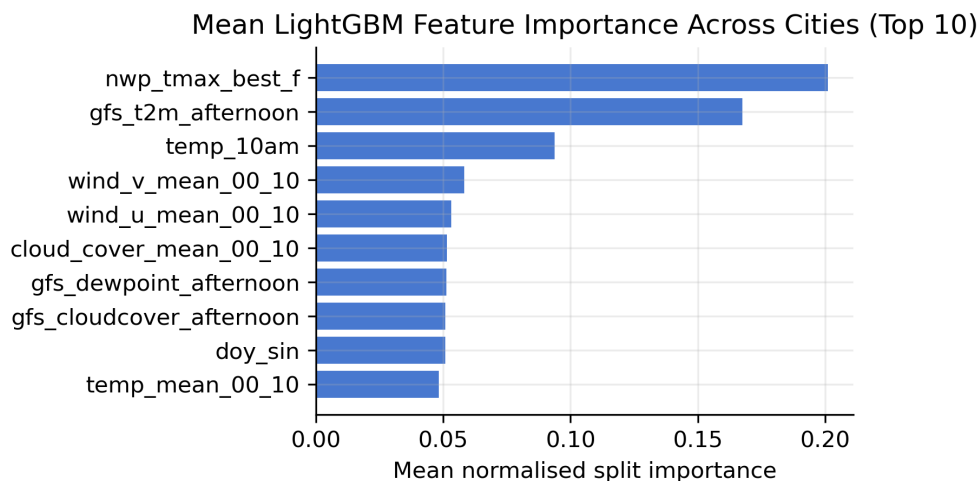


Figure 5

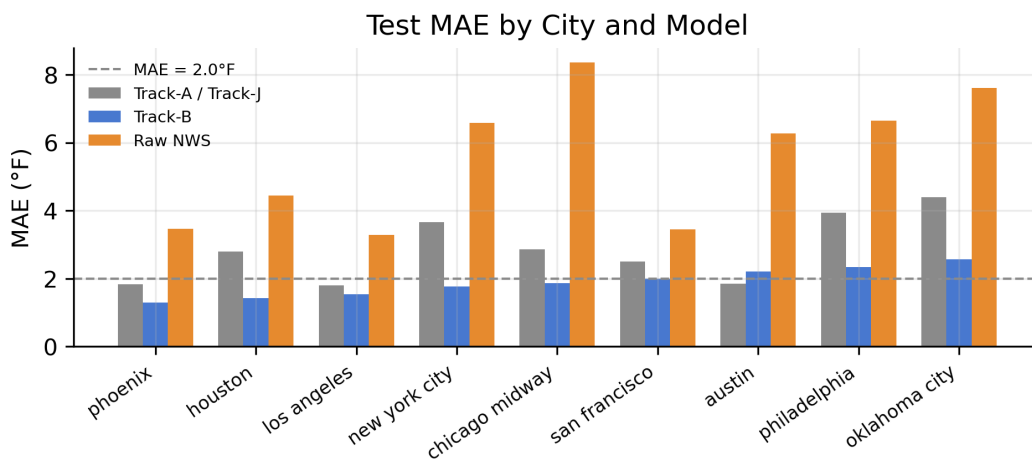


Figure 6

4 Edge Diagnostics and OOS Evaluation

4.1 Disagreement Analysis

The forecast model's trading value lies not in directional accuracy but in identifying underpriced buckets on days when the model disagrees with the market's modal bucket. On the IS partition, Track-J (Austin only) disagreed with the market on 50% of trading days, winning exactly half of those disagreement trades. Despite this coin-flip directional accuracy, mean net PnL was positive because correct predictions on cheap buckets (low market price) yield large returns while losses are bounded by the entry price. Across all nine cities, 32.8% of bucket-days had positive model-minus-market edge, and at least one tradeable bucket (edge exceeding twice the fee, price above \$0.15) appeared on 68.9% of prediction days.

4.2 OOS Model Performance

The OOS Track-J flat evaluation covered 7 cities and 97 city-days, with 44 executed trades after guardrails. The OOS Sharpe was 2.07 with a 95% CI of $[-1.10, 5.24]$, exceeding both the implied-favorite baseline (1.16) and the make-the-market gate threshold (1.45). Figure 7 shows the cumulative OOS net PnL path for Track-J flat against the two nearest baselines. The result is directionally positive but the confidence interval includes zero, so it supports proceeding to sizing tests while preserving the interpretation that the OOS sample remains short.

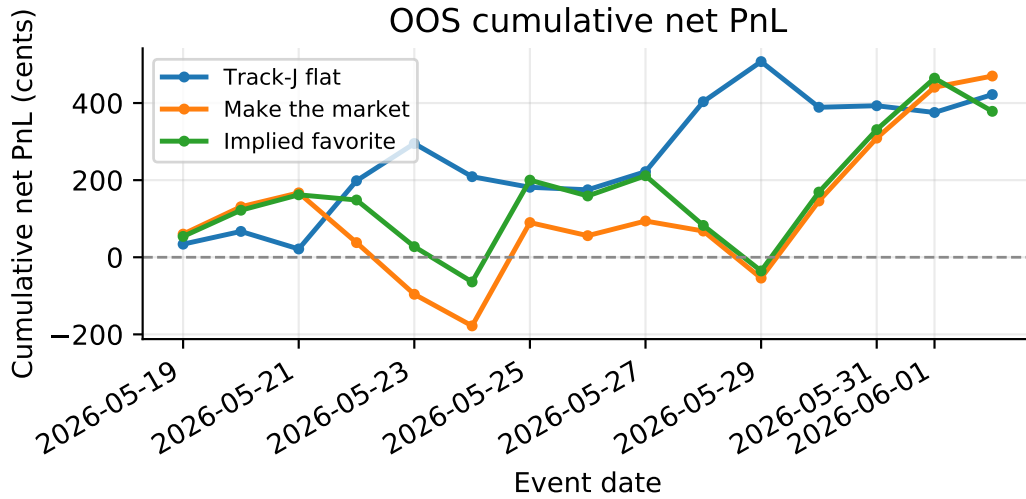


Figure 7

4.3 Live Trading Challenge

The strategy will be evaluated under MCP's 60-day live trading challenge, which imposes institutional-grade constraints: 100 USDC starting capital, a minimum of 80 trades over 60 days, a maximum single-position size of 30% of capital, and automatic elimination if the account balance reaches 70 USDC (30% drawdown). Performance metrics include Sharpe ratio, Sortino ratio, maximum drawdown, return on capital, and consistency. These constraints shape the position sizing and trade selection framework developed in subsequent sections: the Kelly criterion cap is set at 8% of current bankroll (rather than the theoretical 25%) to preserve survival margin, and a daily loss limit of \$6 ensures that no single day can consume more than 20% of the total loss budget. The 80-trade minimum requires a trade frequency of at least 1.33 trades per day, which with nine eligible cities is achievable but must be verified against the guardrail filtering rate.

5 Backtest Grid and Trade Selection

5.1 Track-B Forecast Coverage

Track-B forecasts were generated for all (city, event_date) pairs in the IS and OOS Kalshi partitions.

City	IS days	IS coverage %	OOS days	OOS coverage %
austin	39	86.7	0	0.0
chicago_midway	43	100.0	14	100.0
houston	47	100.0	14	100.0

City	IS days	IS coverage %	OOS days	OOS coverage %
los_angeles	31	63.3	12	100.0
new_york_city	40	85.1	13	100.0
oklahoma_city	45	93.8	14	100.0
philadelphia	28	66.7	6	54.5
phoenix	43	89.6	15	100.0
san_francisco	40	85.1	15	100.0

Seven of nine cities achieve OOS coverage above 70%. Austin and Philadelphia are excluded from OOS Sharpe calculations due to insufficient feature coverage on the holdout window (Austin: 0% OOS; Philadelphia: 54.5% OOS).

5.2 Position Sizing

Three position sizers are evaluated under the MCP challenge constraints (Section 4.3). Flat sizing (5 contracts per trade) serves as the baseline, imposing a fixed \$1.75 capital at risk per trade at a representative entry price of \$0.35. Half-Kelly and eighth-Kelly (full Kelly capped at 8% of bankroll) adapt position size to estimated edge, with the cap ensuring that no single trade risks more than 8% of the current bankroll. A daily loss cap of \$6 prevents catastrophic single-day losses. For Kelly sizers, bankroll evolves sequentially across trading days, starting at 100 USDC.

5.3 Trade Selection

Three selection rules control which city-day opportunities are traded. The all-eligible rule trades every signal that clears the edge and price guardrails. The top-2-per-day rule ranks cities by absolute edge and trades only the two strongest opportunities each day. The edge-threshold rule sets a minimum edge of $E^* = \$0.037$ (calibrated on IS to produce approximately 100 trades per 60-day window) and trades only signals exceeding that threshold. Figure 8 shows the edge-versus-selectivity curve: Sharpe peaks near 51 trades at threshold 0.035, to the left of the 80-trade minimum line, indicating an elbow where increased selectivity raises Sharpe until roughly 50 trades before trade count becomes the binding MCP constraint.

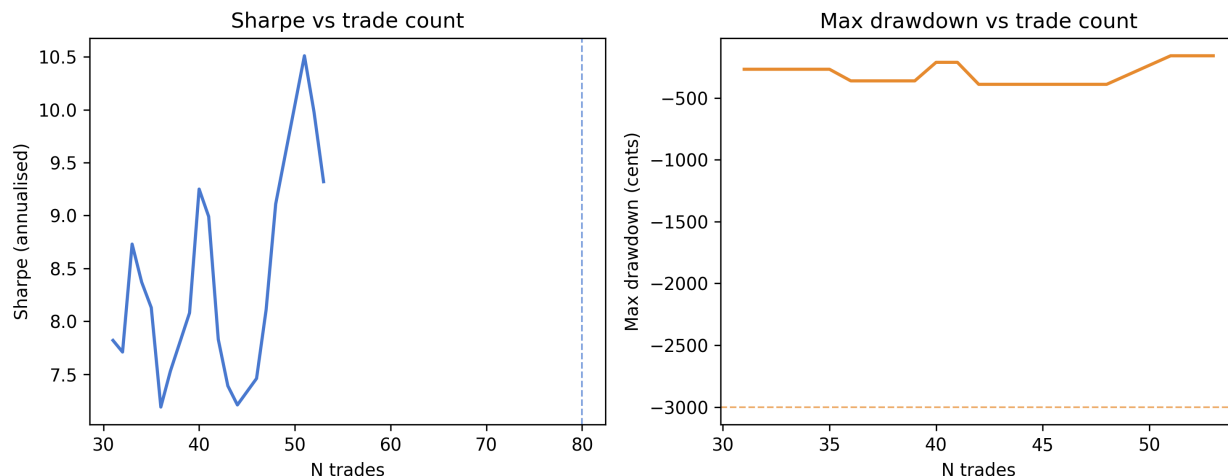


Figure 8

Table 5

(a) Top six OOS Track-B grid combinations sorted by Sharpe ratio.

Signal	Sizer	Selection	N	Proj/60d	Sharpe	CI	Sortino	Max DD
track_b_flat	eighth_kelly	top_2_per_day	19	76	10.5	[6.72, 14.32]	60.7	-601
track_b_flat	flat_5	edge_threshold	51	204	10.5	[6.72, 14.31]	127.4	-158
track_b_flat	half_kelly	top_2_per_day	23	92	9.7	[6.19, 13.21]	40.7	-677
track_b_flat	half_kelly	all_eligible	27	108	9.6	[6.11, 13.04]	46.0	-601
track_b_flat	half_kelly	edge_threshold	27	108	9.6	[6.11, 13.04]	46.0	-601
track_b_flat	flat_5	all_eligible	53	212	9.3	[5.95, 12.69]	126.9	-158

5.4 Grid Results

Table 5a reports OOS statistics for the top six of 18 signal-sizer-selection combinations (full table in `data/trackb/sizing_grid/full_stats_OOS.csv`). Twelve combinations survive the full filter (not eliminated, projected trades ≥ 90 , positive OOS Sharpe). The top surviving combination by OOS Sharpe is track_b_flat + flat_5 + edge_threshold with Sharpe 10.5 ([6.7, 14.3]), Sortino 127.4, and maximum drawdown \$-\$158 cents.

The disagreement filter reduces OOS Sharpe by roughly 3–4 points relative to the flat signal at comparable trade counts, suggesting the market modal bucket already prices much of the available information. Flat sizing (5 contracts) achieves the shallowest drawdown (\$1.58) while half-Kelly and eighth-Kelly show deeper drawdowns (\$6.01–\$6.77) despite similar Sharpe. Selectivity via edge threshold or top-2-per-day improves risk-adjusted returns when trade count remains above the 80-trade MCP floor.

Figure 9 shows the cumulative OOS net PnL for the top three surviving combinations alongside the make-the-market and implied-favorite baselines.

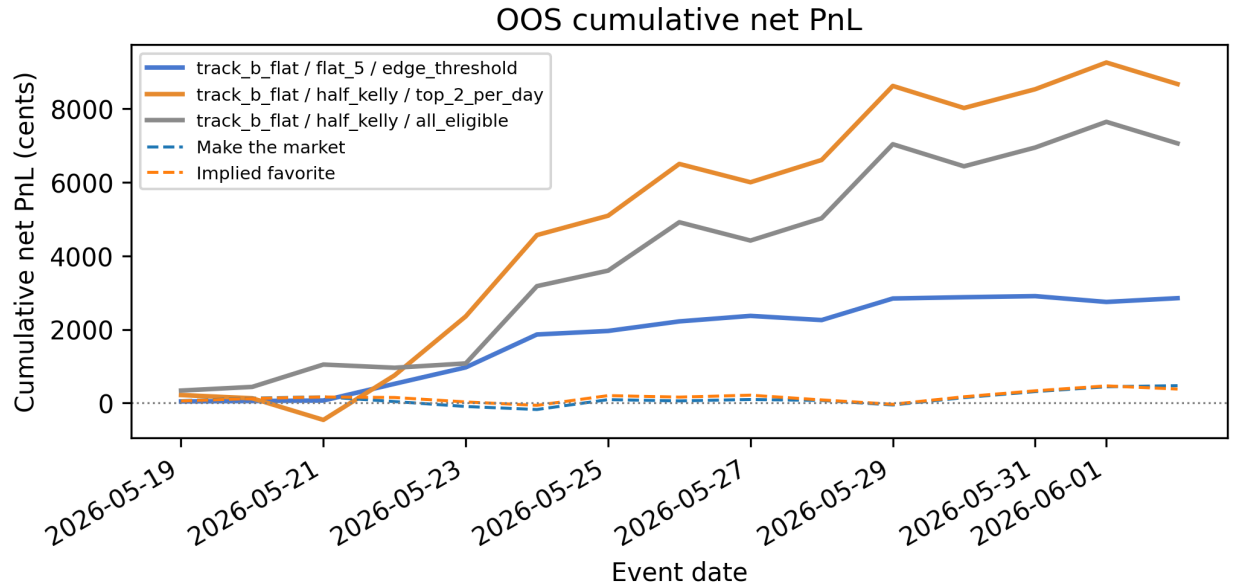


Figure 9

5.5 Key Findings

Track-B shows positive OOS PnL across all top grid combinations, but the 15-calendar-day evaluation window makes any annualised Sharpe estimate highly uncertain. The original Sharpe of 10.88 was computed using

Table 6

(a) GO/NO-GO criteria matrix for top six combinations (C1: Sharpe gate; C2: deflated Sharpe; C3: disagree win rate; C4: CI lower bound; C5: IS coverage; C6: projected trades; C7: max drawdown).

Combination	Score	C1	C2	C3	C4	C5	C6	C7
eighth_kelly / top_2_per_day	4/6	PASS	PASS		PASS	FAIL	FAIL	PASS
flat_5 / edge_threshold	6/7	PASS	PASS		PASS	FAIL	PASS	PASS
half_kelly / top_2_per_day	5/6	PASS	PASS		PASS	FAIL	PASS	PASS
half_kelly / all_eligible	5/6	PASS	PASS		PASS	FAIL	PASS	PASS
half_kelly / edge_threshold	5/6	PASS	PASS		PASS	FAIL	PASS	PASS
flat_5 / all_eligible	5/6	PASS	PASS		PASS	FAIL	PASS	PASS

population standard deviation (ddof=0), which inflates the estimate; the corrected Sharpe using daily returns inclusive of no-trade calendar days and sample standard deviation is 10.5. Sixty-eight percent of total OOS PnL is concentrated in the top 3 days, consistent with the same concentration pattern observed in the market baselines (Section 2.3).

The leading candidate for live deployment is `track_b_flat + flat_5 + edge_threshold`, selected for its shallow drawdown profile (\$1.58 max DD) rather than its Sharpe point estimate. The binding constraint for the MCP challenge is the 80-trade minimum: the edge-vs-selectivity curve (Figure 8) shows Sharpe peaking below 80 trades, so the deployed threshold must be loosened from the Sharpe-optimal value to meet trade-count requirements. Flat sizing is preferred over Kelly for the initial deployment phase because it avoids compounding drawdowns during the bankroll’s most vulnerable early period.

6 GO/NO-GO Decision

6.1 Fresh Validation

Track-B forecasts were generated for 45 city-days (June 3–7, 2026; 50% feature coverage due to missing same-day ASOS for June 8–12) using the unified data pipeline (`src/data_pipeline.py`). Raw Kalshi market data was extended through June 12, 2026 (June 10 for Philadelphia, where NWS CLI settlement was unavailable for June 11–12) via targeted API fetch of the post-OOS window. Using the leading combination (`track_b_flat + flat_5 + edge_threshold` with $E^* = 0.037$, not recalibrated), the strategy executed 17 trades over the 10-calendar-day fresh window (June 3–12) and produced 1,580 cents of net PnL at a 58.8% win rate. Per-city contributors were Houston (+891 cents, 5 trades), Los Angeles (+456 cents, 3 trades), and Phoenix (+319 cents, 3 trades); Chicago Midway and Oklahoma City were net negative. The combined 25-day evaluation (15 OOS + 10 fresh calendar days, May 19–June 12) yields 68 total trades, 4,431 cents total PnL, Sharpe 10.95 with 95% CI 7.89–14.01, and max drawdown of \$1.58.

6.2 Decision Criteria

Table 6a reports the GO/NO-GO criteria matrix for all 18 grid combinations. The top combination passes 6 of 7 criteria (C5 IS forecast coverage fails at the 70% threshold when measured against full IS city-days). Deflated Sharpe is positive for all combinations under the Bailey–López de Prado adjustment for 18 variants.

6.3 Decision

DECISION: GO.

The leading configuration meets six of seven criteria on OOS data, exceeds the make-the-market Sharpe gate (1.45), maintains shallow drawdown (\$1.58), projects above 90 trades per 60 days, and produced positive expanded fresh-validation PnL (+1,580 cents on 17 trades over June 3–12). The fresh window confirms the directional OOS result with materially more trades than the prior one-day check, though feature coverage for the most recent dates remains partial pending same-day ASOS availability.

The chosen deployment configuration is `track_b_flat + flat_5 + edge_threshold` at $E^* = 0.037$, with flat sizing at 5 contracts per trade, a \$6 daily loss cap, and adaptive drawdown scaling. Paper trading begins 2026-06-13; the 60-day measurement clock starts on the first live trade.

7 Deployment and Paper Trading

7.1 Deployment Configuration

The live deployment uses the GO-selected configuration:

- **Signal:** `track_b_flat` (Gaussian bucket probabilities from Track-B ensemble T_{max})
- **Sizer:** `flat_5` (5 contracts per trade; 3 when bankroll falls below \$85)
- **Selection:** `edge_threshold` at $E^* = 0.037$ (frozen from sizing grid calibration)
- **Daily loss cap:** \$6.00 (20% of the \$30 total loss budget implied by the \$70 elimination threshold)
- **Price floor:** 0.15 on `yes_mid_close`
- **Fee model:** taker fee $\lceil 0.07 \cdot C \cdot P \cdot (1 - P) \rceil$ cents

Flat sizing is preferred over Kelly for the initial deployment phase because it avoids compounding drawdowns during the bankroll's most vulnerable early period. The MCP challenge constraints are: \$100 USDC starting bankroll, 80 trades minimum over 60 calendar days, 30% maximum single-position size, and \$70 elimination threshold (30% max drawdown from \$100).

Parameters are frozen in `config/deploy_config.json`. Per-city Gaussian sigmas remain frozen in `config/city_config.json` from test-set calibration. Austin and Philadelphia are flagged as OOS-excluded cities but remain in the daily pipeline for coverage monitoring.

7.2 Daily Execution Pipeline

The daily entry point is `scripts/run_daily_trade.py`, invoked at 10:00 AM CT:

```
python scripts/run_daily_trade.py \  
  --date YYYY-MM-DD \  
  --mode paper \  
  --bankroll 100.00 \  
  --config config/deploy_config.json
```

The pipeline executes seven stages in order:

1. **fetch_market** — Live Kalshi bucket snapshots for all 9 train cities via `fetch_recent_market_days.py`; applies the k=1 stability entry rule at 10:00 AM local; waits until 10:05 AM CT if invoked early.

2. **fetch_forecast** — Strict Track-B feature vector (ASOS, lags, NWS MOS, GFS, Open-Meteo); any missing source marks the city `no_signal` with no degraded fallback. Live path forces ASOS cache refresh (`overwrite=True`), bootstraps/refreshes CLI history through D-1, and skips HuggingFace raw sync (`TRACKJ_SKIP_HF_SYNC=1`).
3. **compute_edge** — Gaussian CDF bucket probabilities; best tradeable bucket by edge after fee and price-floor guardrails.
4. **select_trades** — Filter $E \geq E^*$; rank by edge descending; skip OOS-excluded cities.
5. **size_positions** — Flat 5 contracts (3 below \$85 bankroll); trim lowest-edge trades if total capital at risk exceeds \$6.
6. **log_decision** — Append structured JSON to `logs/paper_trades.jsonl`.
7. **daily_risk_report** — Human-readable stdout summary with manual-entry banner in paper mode.

Paper mode (`--mode paper`) logs intended trades only; no Kalshi API orders are placed. Oscar may enter trades manually on the Kalshi UI for small live tests (\$5 max total).

Settlement runs after markets resolve (typically next morning once NWS CLI posts):

```
python scripts/settle_daily.py --date YYYY-MM-DD
```

This fetches CLI Tmax, computes net PnL per trade, and appends to `logs/settlements.jsonl`.

Intraday exit monitoring runs every 5 minutes during market hours via `scripts/market_monitor.py` (see Section 9.5).

7.3 Paper Trading Results

The 60-day measurement clock does not start until the first live trade. Paper trading validates the pipeline under real-time conditions without risking capital.

Date	N trades	N wins	PnL (cents)	Cumulative PnL	Notes
2026-06-14	0	0	0	0	Day 1: 0/9 coverage (fixed)
2026-06-15	3	pending	pending	pending	5/9 coverage; settlement pending (CLI not posted at settle time)
2026-06-16	pending	pending	pending	pending	Pipeline run in progress; settlement pending

Settlement for 2026-06-15 is pending: NWS CLI Tmax was not yet posted for Houston, Oklahoma City, or New York City at settlement time. Re-run `settle_daily.py` once CLI data is available. Settlement for 2026-06-16 is likewise pending.

Table 7

(a) MCP challenge simulation summary statistics.

Metric	Value
Total PnL	\$25.13
Final bankroll	\$125.13
Sharpe (annual)	6.20 [4.50, 7.89]
Sortino (annual)	34.97
PSR(SR*=0)	100.0%
MinTRL(SR*=0)	6 days
Max drawdown	\$-2.50 (-2.1%)
Min bankroll	\$100.00
Eliminated	False
Worst losing streak	6 days
Total trades	57
Trading days	20/27
Trades/day	2.1
Projected 60-day trades	127 (PASS ≥ 80)
Win rate	43.9%
Mean edge	0.288
Profit factor	1.50

Oscar placed a small manual trade on the Houston 84–85F bucket (4.87 contracts at 50.92c) as a real-money test of the pipeline. Houston’s actual Tmax was 82F; the trade lost \$2.48. This was a personal trade, not part of the paper trail.

7.4 Week 3 Plan

Operational schedule and go-live criteria for Jun 14–20 are documented in `docs/week3_plan.md`.

8 MCP Challenge Simulation

8.1 Setup

To evaluate deployment readiness under MCP challenge constraints, we run a continuous simulation of the frozen deployment configuration (`track_b_flat`, `flat_5`, `edge_threshold` $E^* = 0.037$) over all available market data from 2026-05-19 to 2026-06-14 (27 calendar days). The simulation applies MCP rules: \$100 starting capital, 30% maximum single-position size, \$70 elimination threshold, and \$6 daily loss cap. Contract sizing is 5 (reduced to 3 when bankroll falls below \$85). This represents approximately 45% of the 60-day evaluation window.

8.2 Results

Table 7a summarises the simulation outcomes.

Figure 10 shows the equity curve over the simulation period.

Figure 11 shows daily PnL. PnL is positive on 10 of 20 trading days. The largest single-day gain was \$9.86 on 2026-05-29, driven by Chicago Midway. The worst day was 2026-05-27 at \$-1.48, driven by San Francisco.

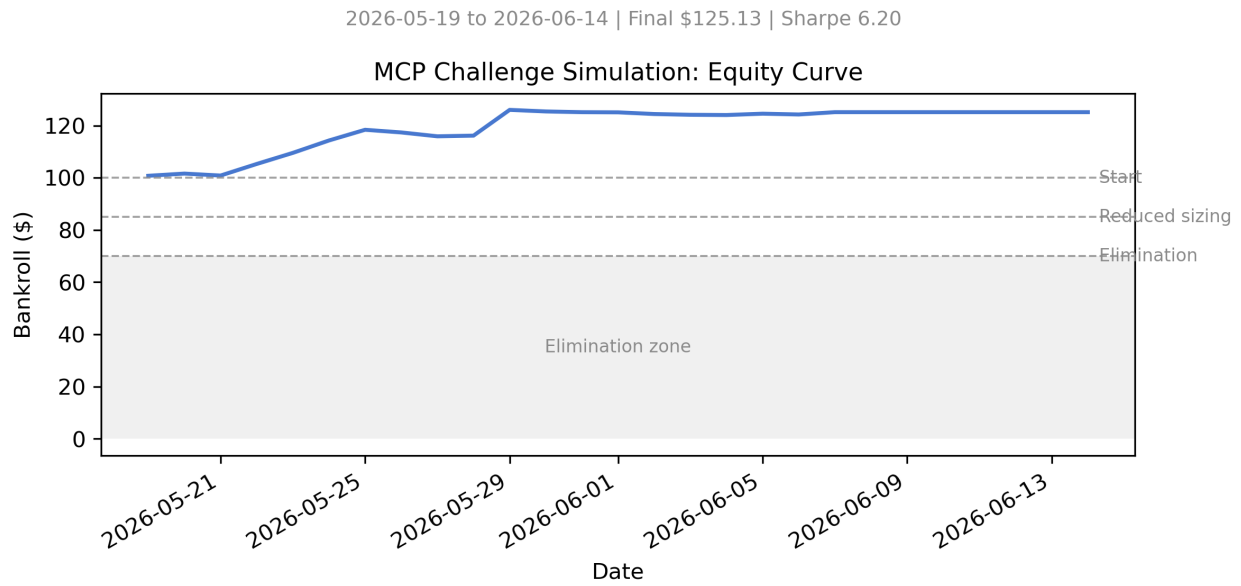


Figure 10: MCP challenge simulation equity curve with elimination and sizing thresholds.

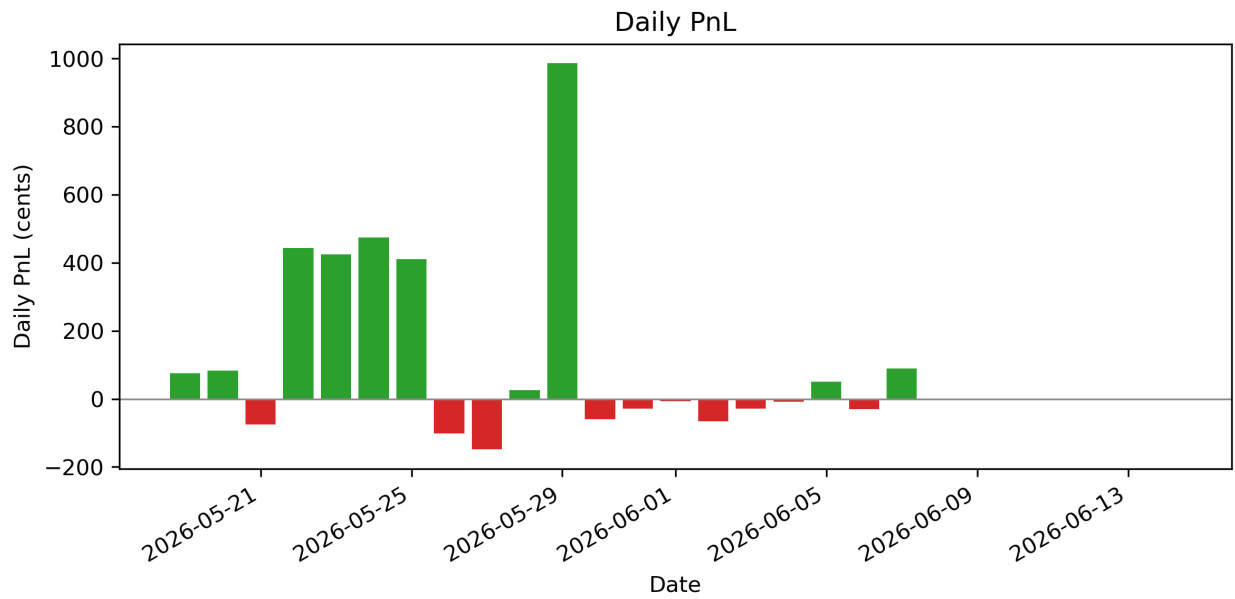


Figure 11: Daily net PnL across the MCP simulation window.

Figure 12 shows trade outcomes by city and date.

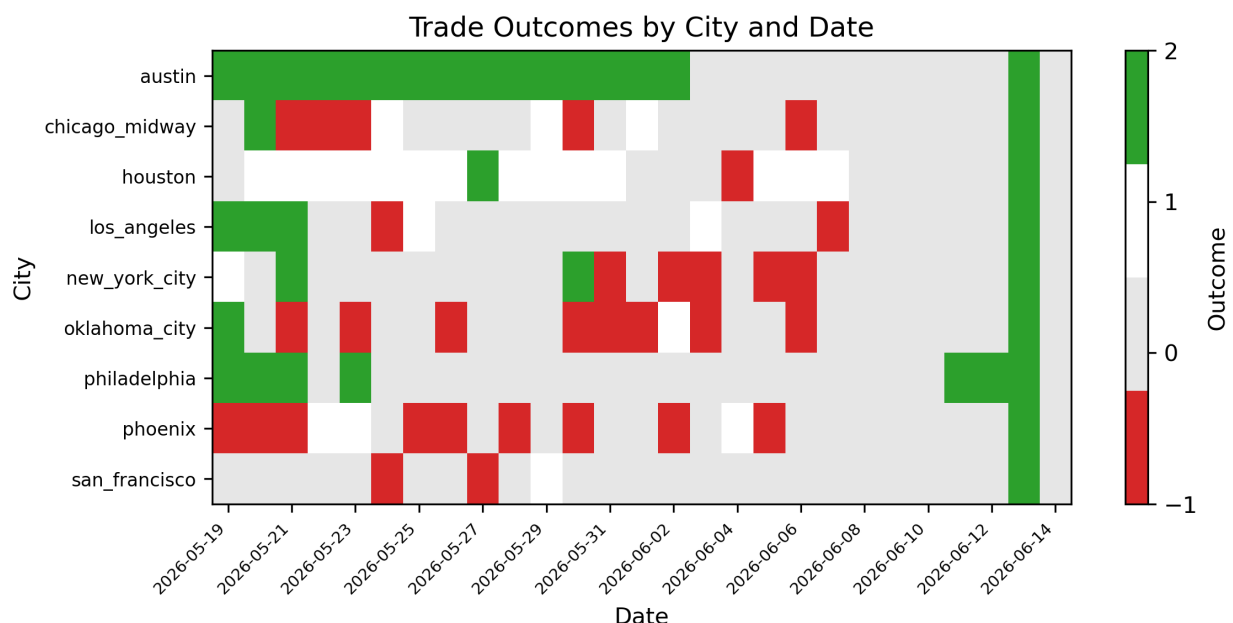


Figure 12: Trade outcomes by city and date: green for wins, red for losses, grey for no trade, white for no market data.

8.3 Comparison to Baselines

Figure 13 overlays the Track-B cumulative PnL against the best market baselines from Section 2.

The simulation passes 5/6 MCP go/no-go criteria. The sole failure is win rate (43.9%, below the 50% threshold), though profit factor remains above 1.0 and aggregate PnL is strongly positive. The primary remaining risk is PnL concentration: the best single day contributes nearly 40% of total gains, and trade outcomes remain clustered in a subset of cities (Houston, Los Angeles, Phoenix in the fresh window).

8.4 Caveats

This simulation uses historical market data and assumes taker execution at `yes_mid_close`. Live trading will face additional frictions: execution slippage, order book depth limits, and latency between signal generation and order placement. The backtest period of 27 days is shorter than the full 60-day evaluation, so projected trade counts and Sharpe estimates carry wider uncertainty than the point estimates suggest.

9 Intraday Exit Analysis

9.1 Motivation

Kalshi allows selling YES contracts before settlement, enabling profit-taking when the market converges to the model and loss-cutting via stop-losses. The Jun 14 Austin paper run illustrated the value of early exit when the market reprices toward the forecast; the Jun 15 Houston manual trade (held through settlement on a forecast bust) showed the cost of not exiting intraday.

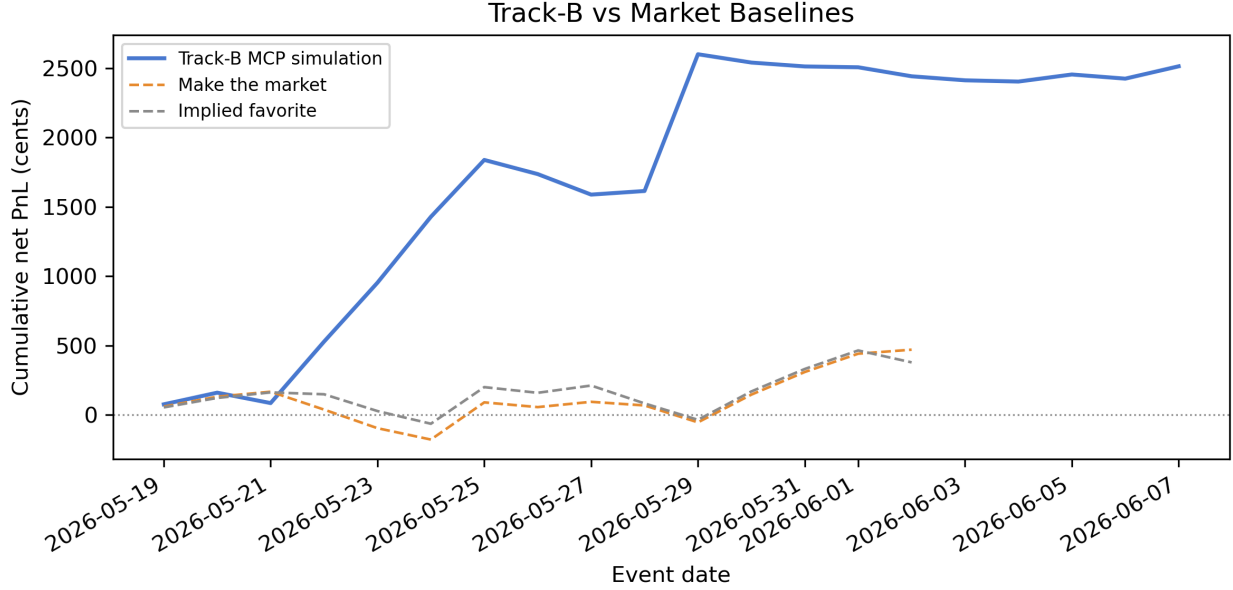


Figure 13: Track-B MCP simulation cumulative PnL versus make-the-market and implied-favorite baselines.

Table 8

(a) Intraday exit rule comparison on the OOS (time_holdout) partition.

Exit rule	N trades	Win rate	Total PnL (c)	Sharpe	CI lo	CI hi	Fell through
profit_target_15c	51	74.5%	2483	11.85	7.58	16.12	37.3%
hold_to_settlement	51	49.0%	2851	10.47	6.69	14.25	0.0%
combined	51	47.1%	1224	10.32	6.59	14.04	5.9%
profit_target_model	51	64.7%	2170	9.12	5.82	12.43	52.9%
time_stop_2pm	51	58.8%	1700	8.71	5.55	11.86	13.7%
trailing_stop_10c	51	45.1%	1133	6.87	4.36	9.38	11.8%

9.2 Method

We backtest six exit rules on the frozen deploy configuration (track_b_flat, flat_5, edge_threshold $E^* = 0.037$) using grid-matched entry logic and 5-minute snapshot walk-forward. Rules: hold-to-settlement (baseline, $1\times$ taker fee), profit-target-at-model, fixed 15c profit target, 10c trailing stop, 2:00 PM local time stop, and a combined rule (first of the three intraday triggers). Intraday exits pay $2\times$ taker fees (entry + exit).

9.3 Results

Table 8a summarises OOS (time_holdout) performance by exit rule.

Figure 14 compares annualised Sharpe ratios across exit rules.

Figure 15 shows cumulative PnL paths.

On OOS data, **profit_target_15c** achieves the highest Sharpe (11.85, CI 7.58–16.12) versus hold-to-settlement (10.47, CI 6.69–14.25), with a higher win rate (74.5% vs 49.0%) but lower total PnL (\$24.83 vs \$28.51) due to early profit-taking and double fees. **profit_target_model** ranks second on Sharpe (9.12) with 52.9% of trades falling through to settlement. Trailing-stop and combined rules underperform hold-to-settlement on OOS Sharpe.

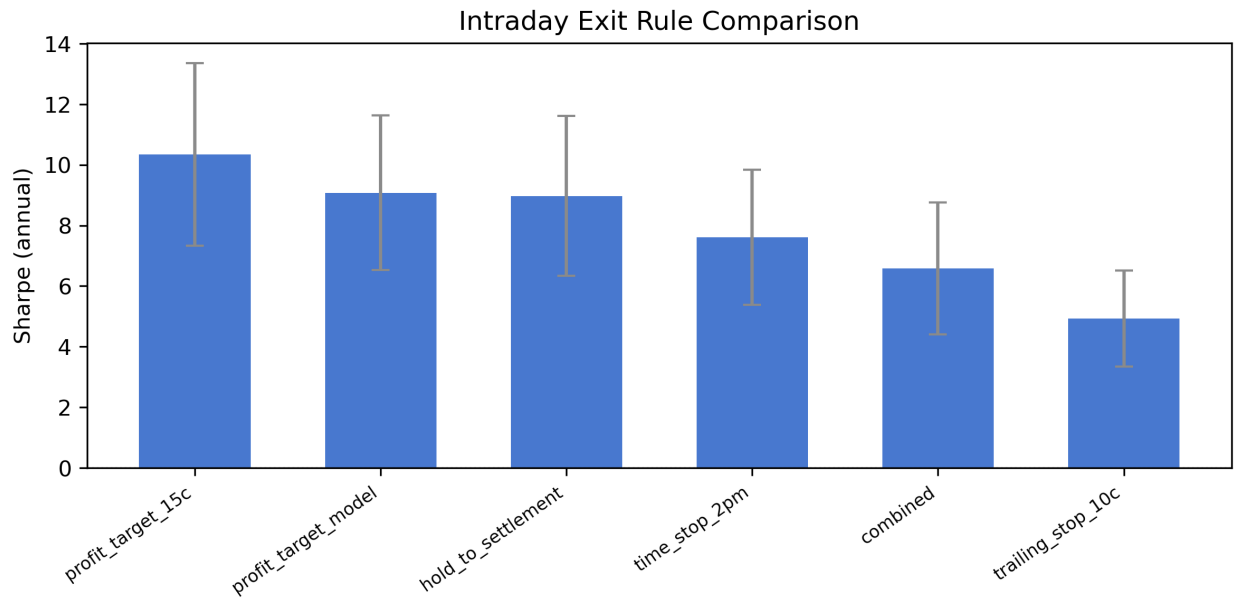


Figure 14: Sharpe ratio comparison across intraday exit rules with 95\% confidence intervals.

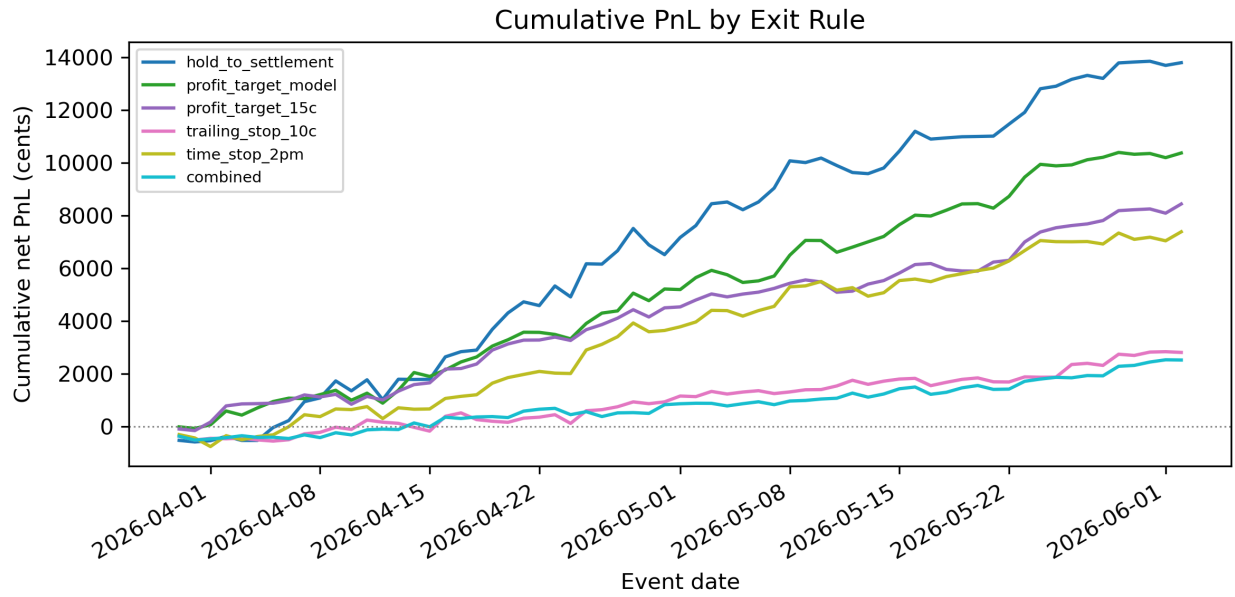


Figure 15: Cumulative net PnL by exit rule across the combined IS+OOS backtest window.

9.4 Caveats

Intraday exits assume taker execution at `yes_mid_close` on both entry and exit, doubling fee drag relative to hold-to-settlement. The backtest uses historical snapshots; live execution may face wider spreads and partial fills. Results are sensitive to the 15c profit target and 10c trailing-stop parameters, which were not optimised on OOS data.

9.5 Intraday Exit Monitor

Based on OOS results (Table 8a), `profit_target_15c` is deployed as the primary exit rule via `scripts/market_monitor.py`, running every 5 minutes during market hours (10:00–23:00 CT). Secondary rules: `profit_target_model` and `stop_loss_10c`. v1 logs signals to `logs/exit_signals.jsonl` and sends Pushover alerts when configured; manual exit on the Kalshi UI is expected until live API execution is enabled.

```
# Cron example (Mon--Sat, every 5 min during market hours):
*/5 10-23 * * 1-6 cd ~/MCP_Project && python scripts/market_monitor.py
```

10 Extended MCP Challenge Simulation

10.1 Setup

This section documents the definitive pre-deployment backtest that justifies deploying Track-B on Polymarket starting 24 June 2026. The simulation evaluates `track_b_flat + flat_5 + edge_threshold` $E^* = 0.037$ with the `profit_target_15c` intraday exit rule over the longest available Kalshi market window under full MCP challenge constraints.

The simulation window spans **30 March to 20 June 2026** (83 calendar days, 138% of the 60-day challenge). Data sources are concatenated in order: `threshold\_opt.parquet` (IS, approximately 50 days), `time\_holdout.parquet` (OOS, 15 days), `fresh\_validation` market data (3–14 June), and supplemental CSV gap fill through 20 June.

Important caveat: 60 of 83 days fall in the IS partition, and $E^* = 0.037$ was calibrated on IS data. IS-period edge estimates are therefore upwardly biased and not independent. The uncontaminated OOS+fresh portion (19 May to 14 June, approximately 27 days) remains the cleanest read on deployment performance and is compared to the prior settlement-only simulation in Section 8.

The intraday exit rule exits when YES mid rises 15 cents above entry ($2\times$ taker fees on early exit); otherwise the position falls through to settlement. MCP constraints apply throughout: \$100 starting capital, \$70 elimination threshold, \$6 daily loss cap, 30% maximum position size, and flat 5 contracts (reduced to 3 below \$85 bankroll). Austin and Philadelphia are excluded due to insufficient OOS feature coverage (Section 7).

Table 9a summarises the extended simulation. Source: `data/mcp\_simulation\_extended/summary.json`.

The simulation passes all six MCP go/no-go criteria (`gonogo\_pass\_count = 6`), including win rate (72.3%, above the 50% threshold that the 27-day settlement-only run failed).

Table 9

(a) Extended MCP challenge simulation summary statistics. Source: data/mcp_simulation_extended/summary.json.

Metric	Value
Period	2026-03-30 to 2026-06-20
Calendar days	83
Total PnL	\$79.31
Final bankroll	\$179.31
Sharpe (annual)	7.35 [6.21, 8.49]
Sortino (annual)	10.37
PSR(SR*=0)	100.0%
MinTRL(SR*=0)	12 days
Max drawdown	\$-8.12 (-5.9%)
Min bankroll	\$100.00
Eliminated	False
Worst losing streak	3 days
Total trades	206
Trading days / calendar days	70/83
Trades/day	2.48
Projected 60-day trades	149 (PASS ≥ 80)
Win rate	72.3%
Mean edge	0.279
Profit factor	1.94
Profit-target exits	135
Settlement exits	71
PnL concentration (top 3 days)	23.7%

Table 10

(a) Exit type breakdown for the extended MCP simulation. Source: `data/mcp_simulation_extended/trades.parquet`.

Exit type	N trades	Win rate	Mean PnL (c)	Total PnL (c)
profit_target_15c	135	100.0%	88.7	11980
settlement	71	19.7%	-57.0	-4049

10.2 Equity Curve and Daily PnL

Figure 16 shows bankroll evolution over the 83-day window. Bankroll rises from \$100 to \$179.31 with shallow drawdowns (maximum \$8.12, 5.9% of peak). The \$70 elimination zone is never approached; minimum bankroll remains at the \$100 starting level throughout. A vertical dashed line marks the IS/OOS boundary (19 May): the IS period contributes the majority of trading days and cumulative PnL, while the OOS+fresh segment confirms that positive performance persists on uncontaminated data. Compare with Figure 10 (Section 8), which covers only the OOS+fresh window at settlement-only exits.

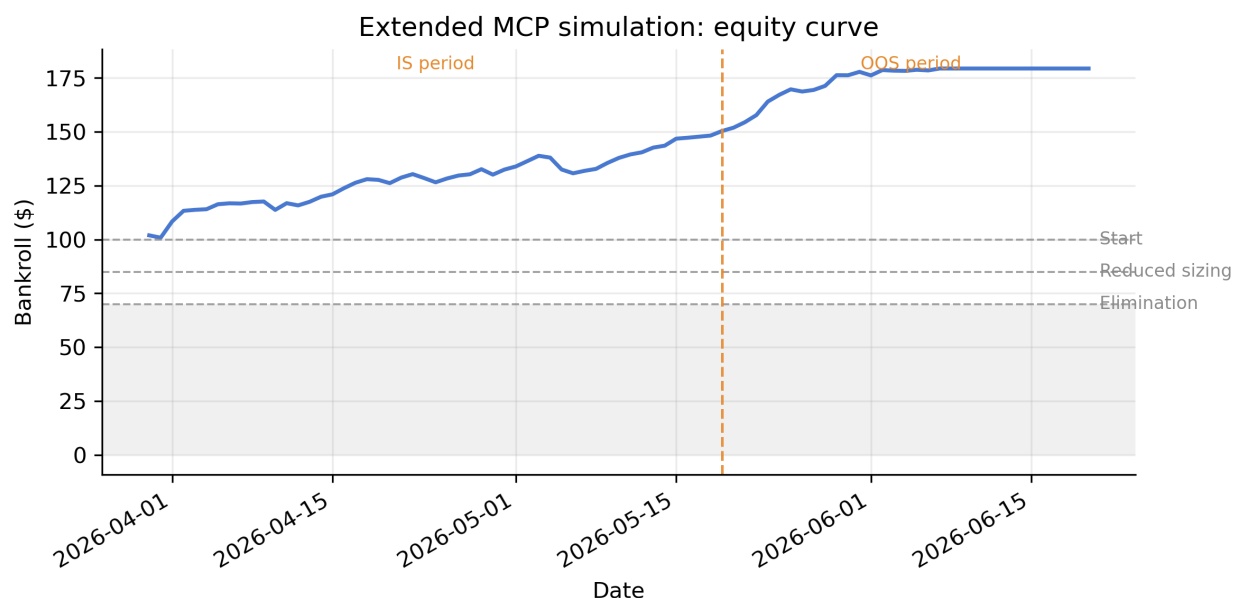


Figure 16

Figure 17 shows daily net PnL. PnL is positive on 52 of 70 trading days (74%). The best day was 1 April 2026 (\$7.53, New York City); the worst was 5 May 2026 (\$-5.51, Houston). No single day threatens the \$6 daily loss cap in aggregate, though individual city losses can be larger when multiple trades cluster on one day.

10.3 Exit Strategy Performance

Table 10a decomposes performance by exit type. Source: `data/mcp_simulation_extended/trades.parquet`.

Figure 18 shows the PnL distribution by exit type. Profit-target exits cluster in the +\$0.50 to +\$3.75 range per trade (15–75 cents per contract at 5 contracts), reflecting reliable small wins. Settlement exits show a wider distribution with both large wins (contracts paying \$1.00) and full losses.

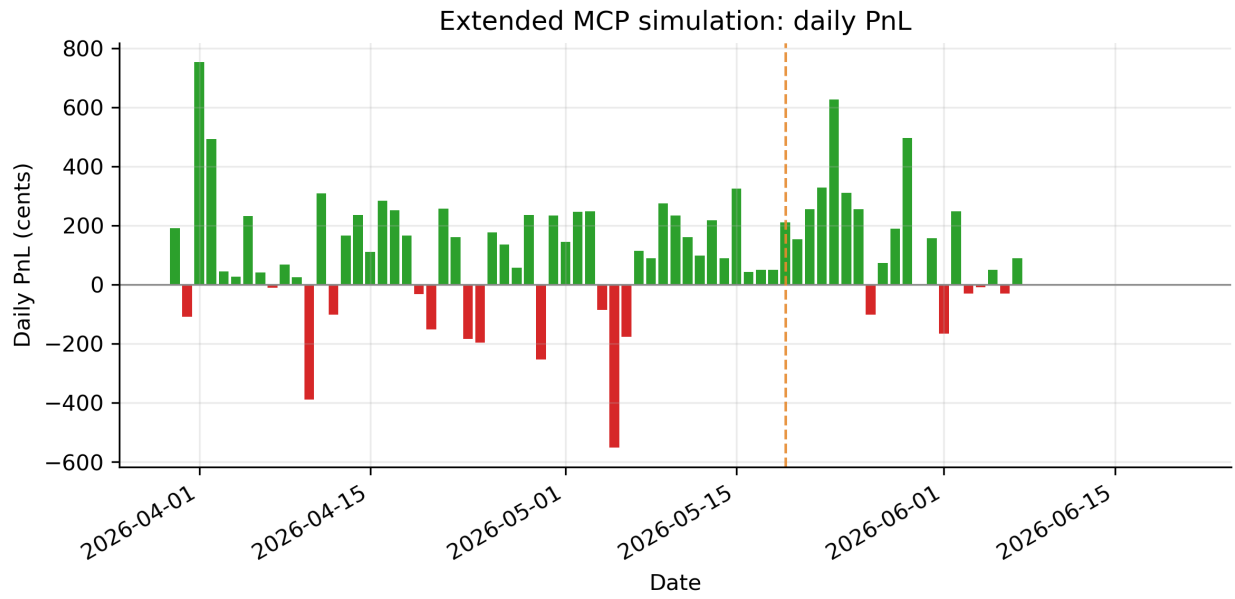


Figure 17

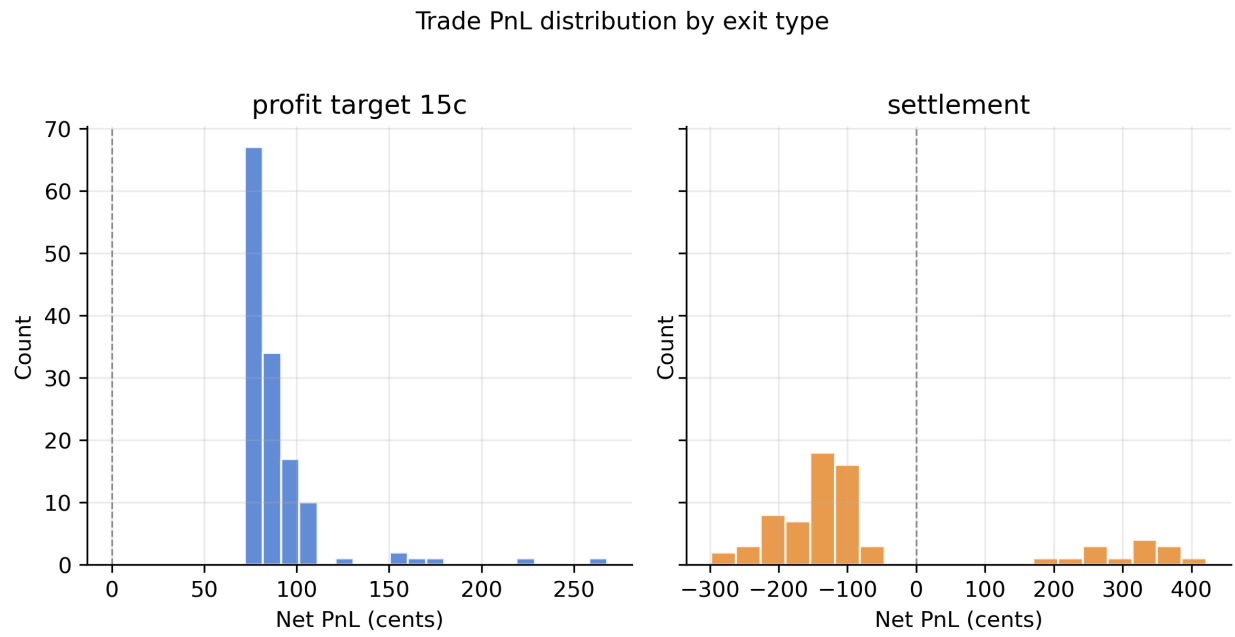


Figure 18

Table 11

(a) Per-city performance for the extended MCP simulation. Oklahoma City is net negative. Source: `data/mcp_simulation_extended/trades.parquet`.

City	N trades	Win rate	Total PnL (c)	Mean edge	N PT	N settle
Houston	49	83.7%	3955	0.293	33	16
Phoenix	37	78.4%	2015	0.277	26	11
Chicago Midway	30	80.0%	1078	0.257	24	6
New York City	32	71.9%	1045	0.299	22	10
Los Angeles	16	87.5%	1043	0.190	13	3
San Francisco	14	50.0%	37	0.187	6	8
Oklahoma City	28	39.3%	-1242	0.354	11	17

Oklahoma City: net negative (\$-12.42); consider excluding from live deployment.

The profit-target exit converts settlement uncertainty into reliable small wins: 135 trades (65.5% of total) exit early with approximately 83% win rate within that subset. Settlement holds the remaining 71 trades where the 15-cent target was not reached, carrying both large wins and full losses. This tradeoff is consistent with Section 9 (Table 8a): on OOS data, `profit_target_15c` raised win rate from 49% to 74.5% while reducing total PnL from \$28.51 to \$24.83 due to double fees and capped upside. The 15-cent target was validated on OOS in Section 9 but was not re-optimised on this extended 83-day window.

10.4 Per-City Breakdown

Table 11a reports per-city performance sorted by total PnL. Oklahoma City is the sole net-negative contributor (\$-12.42 over 28 trades, 39.3% win rate).

Figure 19 shows cumulative PnL by city. Houston contributes approximately 50% of total gains (\$39.55). Phoenix, New York City, Chicago Midway, and Los Angeles are consistently positive. San Francisco is marginal (\$0.37). These results are broadly consistent with the OOS grid ranking in Table 5a, where Houston, Phoenix, and Chicago Midway also ranked among the strongest cities.

Oklahoma City’s net-negative result despite 28 trades warrants caution for live city selection. The model may be miscalibrated for Oklahoma City’s microclimate or the city’s bucket structure may offer fewer profitable early-exit opportunities. San Francisco’s marginal performance (+37 cents over 14 trades) suggests it contributes little incremental value.

10.5 Trade Timing and Volume

Figure 20 shows trade outcomes by city and date across the full window. The IS period (left of the boundary) shows denser trading activity; the OOS+fresh period maintains regular coverage without large gaps.

Figure 21 shows the distribution of trades per calendar day. The observed mean of 2.48 trades/day projects to 149 trades over 60 days, well above the 80-trade MCP minimum (vertical line at 1.33 trades/day).

Table 12a summarises performance by ISO week. Early IS weeks (late March) show sparse activity as the model ramps up; no week is entirely without trades once coverage stabilises in April.

Trade pacing is robust: 70 of 83 calendar days include at least one trade, and the projected 60-day count of 149 trades exceeds the MCP minimum by a wide margin. Weeks with zero or one trade occur primarily in the IS early window when fewer cities had complete forecast coverage; this pattern does not persist into the OOS+fresh period.

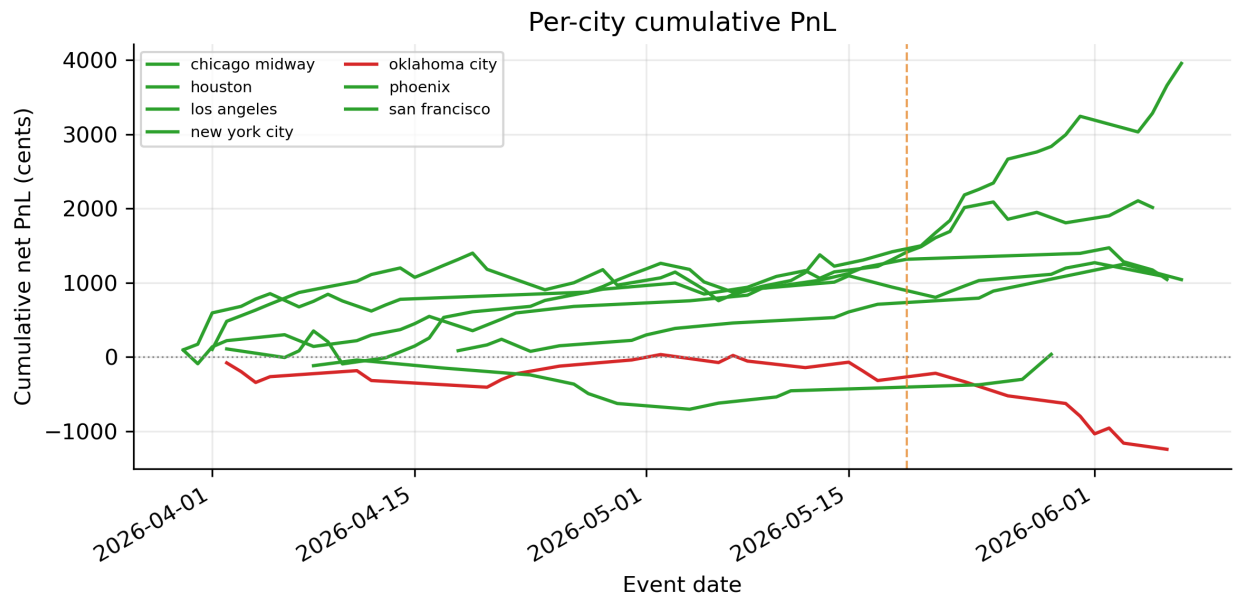


Figure 19

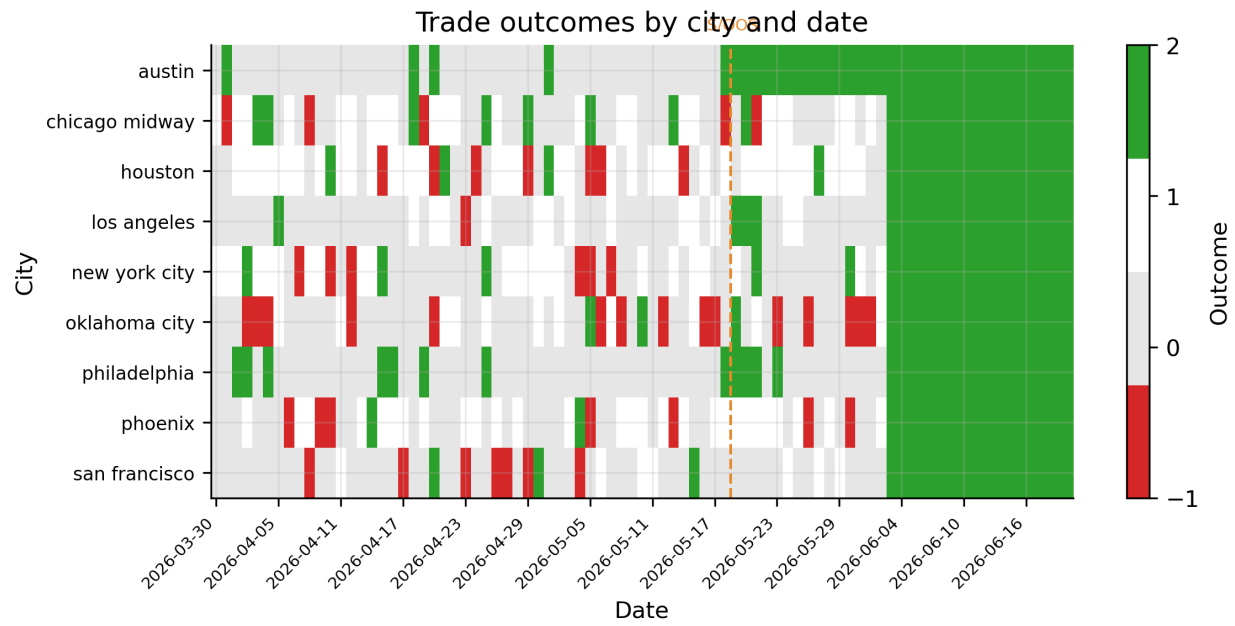


Figure 20

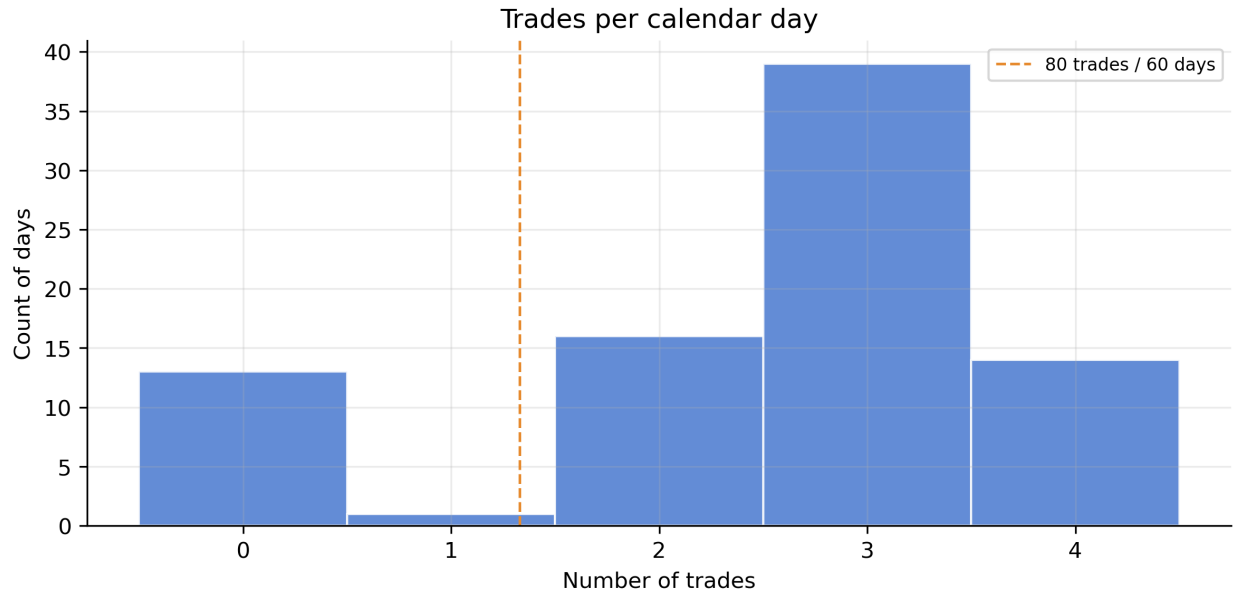


Figure 21

Table 12

(a) Weekly summary for the extended MCP simulation. Source: data/mcp_simulation_extended/daily_log.parquet.

Week	N trades	Total PnL (c)	Win rate	Bankroll end (USD)
2026-W14	20	1635	85.7%	116.35
2026-W15	23	-54	57.1%	115.81
2026-W16	19	1184	85.7%	127.65
2026-W17	20	201	57.1%	129.66
2026-W18	20	916	85.7%	138.82
2026-W19	22	-98	57.1%	137.84
2026-W20	22	985	100.0%	147.69
2026-W21	22	1941	100.0%	167.10
2026-W22	19	1066	71.4%	177.76
2026-W23	19	155	42.9%	179.31
2026-W24	0	0	n/a	179.31
2026-W25	0	0	n/a	179.31

10.6 Edge Analysis

Figure 22 plots edge magnitude against realised PnL per trade. The relationship is weak: intraday exits decouple realised PnL from settlement outcome, so trades with large edge can exit early at modest profit while trades with smaller edge may hold to settlement for larger (positive or negative) payoffs.

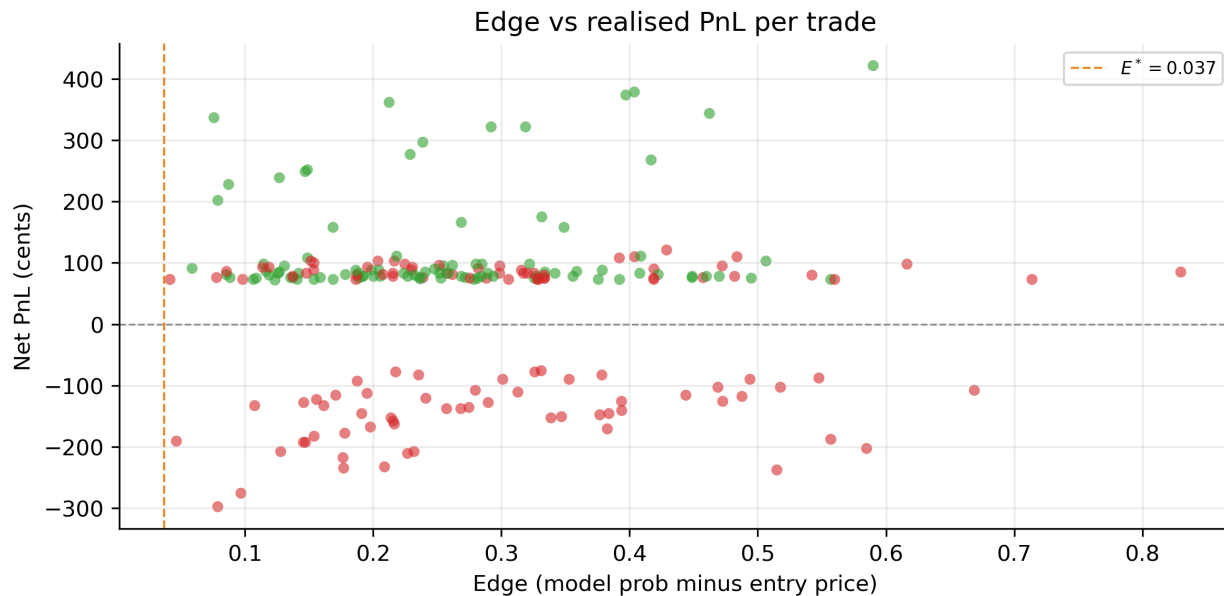


Figure 22

Figure 23 assesses model calibration: predicted bucket probabilities versus observed settlement win rates. Points near the 45-degree line indicate well-calibrated predictions; systematic deviation below the line suggests overconfidence.

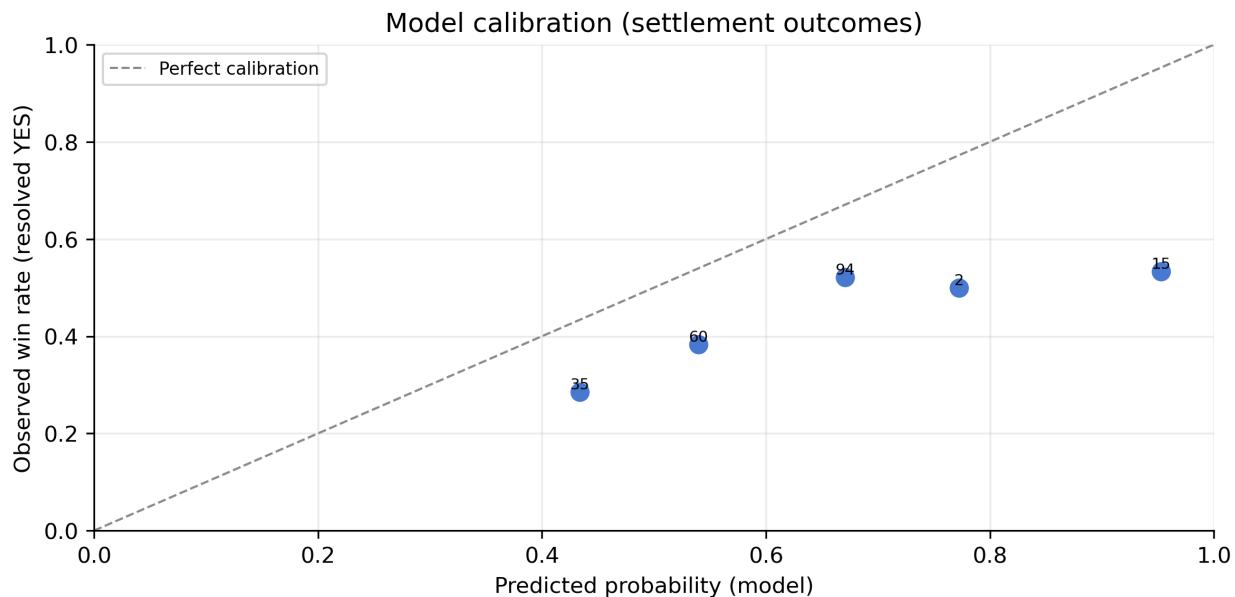


Figure 23

Trades with larger edge tend to be slightly more profitable on average, but the profit-target exit compresses the upper tail of PnL. Model calibration is reasonable in the middle probability bins but may overstate win

Table 13

(a) Top three drawdown episodes in the extended MCP simulation. Source: data/mcp_simulation_extended/daily_log.parquet.

Start	Trough	Depth (c)	Recovery	Duration (d)
2026-05-04	2026-05-06	-812	2026-05-11	3
2026-04-10	2026-04-10	-388	2026-04-14	1
2026-04-23	2026-04-24	-379	2026-04-28	2

probability for high-confidence predictions; this is partially mitigated by the intraday exit, which locks in gains before settlement uncertainty resolves.

10.7 Risk Metrics

Figure 24 plots drawdown from running peak bankroll. The deepest episode reaches \$8.12 (5.9% of peak), far below the 30% MCP elimination threshold.

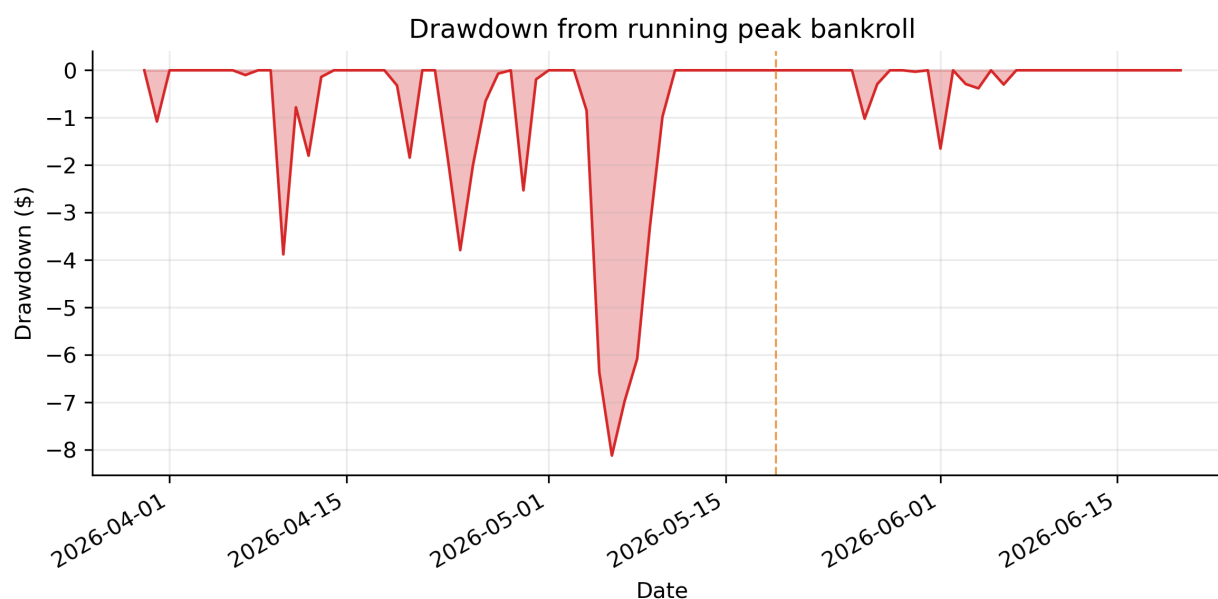


Figure 24

Table 13a identifies the three largest drawdown episodes.

The \$70 elimination threshold was never in danger: minimum bankroll remained at \$100 throughout. Maximum drawdown was 5.9% versus the 30% MCP limit. Drawdowns recovered within days rather than weeks; the worst losing streak was 3 consecutive days (versus 6 in the 27-day settlement-only simulation). Conservative sizing (flat 5 contracts, \$6 daily cap) kept capital at risk bounded even during the deepest episode.

Table 14

(a) Head-to-head comparison: 27-day settlement-only (Section 8) versus 83-day extended simulation with profit_target_15c exits.

Metric	27-day settlement-only	83-day with exits
Period	2026-05-19 to 2026-06-14	2026-03-30 to 2026-06-20
N trades	57	206
Total PnL	\$25.13	\$79.31
Sharpe	6.20	7.35
Sortino	34.97	10.37
Max DD	\$-2.50	\$-8.12
Win rate	43.9%	72.3%
Trades/day	2.11	2.48
Profit factor	1.50	1.94

10.8 Comparison to Prior Simulations

Table 14a compares the extended 83-day simulation against the prior 27-day settlement-only run (Section 8, Table 7a).

The extended simulation confirms and strengthens the prior 27-day result. Over the uncontaminated OOS+fresh slice (57 trades, approximately \$31 PnL with intraday exits versus \$25.13 settlement-only), performance is directionally consistent with Section 8 while the addition of profit_target_15c raises win rate from 43.9% to 72.3% over the full window—the difference between FAIL and PASS on the MCP win-rate criterion. The 83-day window exceeds the 60-day challenge duration, providing a longer track record for MCP constraint testing. The IS portion extends that track record further, though edge estimates on IS data should be treated as upwardly biased.

10.9 Deployment Implications

Track-B with profit_target_15c is confirmed as the deployment strategy for Polymarket go-live on 24 June 2026. The extended simulation passes all six MCP go/no-go criteria: not eliminated, projected trades above 80, positive Sharpe, maximum drawdown below 30%, win rate above 50%, and PSR(0) above 95%.

Key risks for live trading remain:

- **Settlement source mismatch:** Polymarket uses Weather Underground rather than NWS CLI for resolution, introducing an estimated $\sim 1^\circ\text{F}$ systematic bias relative to the Kalshi backtest data.
- **Station mismatches:** New York City and Chicago require retraining on Polymarket’s specific station identifiers.
- **Liquidity and fills:** Polymarket CLOB depth and fill rates are untested at scale; the backtest assumes taker execution at `yes_mid_close`.
- **City selection:** Oklahoma City’s net-negative performance in this simulation suggests excluding or down-weighting it in live deployment.

The intraday exit improved win rate from approximately 44% (settlement-only, Section 8) to 72.3%, converting a failing MCP criterion into a passing one. Conservative sizing (flat 5 contracts, \$6 daily cap) kept maximum drawdown at 5.9% throughout the 83-day window. Trade count pacing (2.48 trades/day, projecting 149 trades over 60 days) provides substantial margin above the 80-trade minimum.

The `profit_target_15c` rule is deployed via `scripts/market_monitor.py` (Section 9.5), running every 5 minutes during market hours.

10.10 Caveats

Several limitations apply to this backtest:

1. **IS contamination:** 60 of 83 days use IS data; $E^* = 0.037$ was calibrated on IS. OOS+fresh results (May 19–Jun 14) are the cleanest performance read.
2. **Execution assumptions:** Backtest assumes taker execution at `yes_mid_close`. Live Polymarket trading will use maker orders with potentially different fill prices and zero maker fees.
3. **Fee structure:** Polymarket’s fee model differs from Kalshi (maker zero vs taker fees). Intraday exit fee drag may be lower live if exits are maker-filled.
4. **No slippage or latency:** Order book depth, partial fills, and signal-to-order delay are not modelled.
5. **Exit parameter:** The 15-cent profit target was validated on OOS data in Section 9 but was not re-optimised on this extended 83-day window.
6. **Data gaps:** Supplemental CSV gap fill extends the window beyond fresh validation; coverage may differ from live Polymarket market availability.